REPRESENTABILITY BY ALGEBRAIC SPACES CHARACTERIZED BY FAITHFULNESS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span. The fibre-product input is not carded (its proof is a citation plus a formal one-liner) and is declared external, together with the faithfulness criterion.

Allowed standard prerequisites (editorial).

- Faithfulness criterion. lemma-criterion-map-representable-spaces-fibred-in-groupoids (same chapter, NOT carded, declared external). Gives (3) implies (2).
- Fibred-category equivalence properties. Categories, lemma equivalence-fibred-categories (cross-chapter, named). Gives (1) iff (2).
- 2-fibre products with representable targets. lemma-2-fibre-product (same chapter, NOT carded, declared external). Makes W = (Sch/U) x_Y X an algebraic stack.
- No-automorphisms proposition. proposition-algebraic-stack-no-automorphisms (carded L26684). Represents the setoid-valued W by an algebraic space.

Source reference. Original reference 003Z: focused-seeds/geometry/S9-alg-characterize-representable-by-algebraic-spaces/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04T2: sources/stacks-algebraic/pinned-original-source.tex, source lines 1964–1964. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02ZY: sources/stacks-algebraic/pinned-original-source.tex, source lines 769–769. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 04SZ: sources/stacks-algebraic/pinned-original-source.tex, source lines 1771–1771. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $f : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of algebraic stacks over S. The following are equivalent:*

1. *for $U \in \mathrm{Ob}((Sch/S)_{fppf})$ the functor $f : \mathcal{X}_U \to \mathcal{Y}_U$ is faithful,*
2. *the functor f is faithful, and*
3. *f is representable by algebraic spaces.*

Proof. Parts (1) and (2) are equivalent by general properties of 1-morphisms of categories fibred in groupoids, see Categories, Lemma 003Z. We see that (3) implies (2) by Lemma 02ZY. Finally, assume (2). Let U be a scheme. Let $y \in \mathrm{Ob}(\mathcal{Y}_U)$. We have to prove that

$$\mathcal{W} = (Sch/U)_{fppf} \times_{y,\mathcal{Y}} \mathcal{X}$$

is representable by an algebraic space over U. Since $(Sch/U)_{fppf}$ is an algebraic stack we see from Lemma 04T2 that $\mathcal{W}$ is an algebraic stack. On the other hand the explicit description of objects of $\mathcal{W}$ as triples $(V, x, \alpha : y(V) \to f(x))$ and the fact that f is faithful, shows that the fibre categories of $\mathcal{W}$ are setoids. Hence Proposition 04SZ guarantees that $\mathcal{W}$ is representable by an algebraic space. □